# The 2026 US–Iran War: A Game-Theoretic Analysis and Forecast Audit

*What happened from February to October 2026, why, what a game-theoretic forecasting engine got right and wrong, and what comes next*

Oct 29, 2026 · @Faye

## Abstract

The 2026 US–Iran war has not climbed an escalation ladder. Since July it has settled into a frozen tanker war: no deal, no full air campaign, and a Strait of Hormuz kept largely shut as leverage. A game-theoretic forecasting engine read the war's physical and structural state correctly but could not predict its principals' decisions.

This report has three parts:

- **Chronology and structure.** It reconstructs the war from February to October 2026 from dated, sourced reporting. It then models the conflict as a *constitutive* dispute over control of a chokepoint: a Fearon commitment problem in which a pause without a deal is cheap for both sides, so stasis, not settlement, is the equilibrium.
- **The forecast record.** The engine's 19 resolved geopolitical forecasts scored 12 correct (63%), Brier 0.236.
  - Calls on physical and structural state went **9 for 9** (Brier 0.13): Hormuz transit, regime survival, the new Supreme Leader's visibility, proxy behavior, nuclear-program reporting.
  - Calls on a principal's discrete decision (whether to strike, stop, sign or cut output) went **1 for 7** (Brier 0.38).
- **Ceasefire timing.** Survival models on 401 historical ceasefires rank collapse risk well (C-index 0.83). They predict the *day* of collapse only when a ceasefire fails within three days.

As of 29 October the engine's live geopolitical calls are on track: no US strike on Iranian territory before the 3 November midterms, and a federal fight over certifying a Senate result. The modal path to 30 November is the continued tanker war (0.55), against a post-midterm US strike campaign (0.35) and a confirmed deal (0.10). The central lesson: forecasting wars means forecasting constraints, and treating single decisions as near-unforecastable.

# 1. Introduction

A strategic reading of a war is only as good as the dated, falsifiable predictions it yields. This report audits one such reading of the 2026 US–Iran war against what actually happened.

The reading comes from GT-Analyst, an engine that turns news into scored forecasts through layered game-theoretic doctrine: chessboard actor decomposition, Fearon's bargaining model, Schelling's commitment and brinkmanship, escalation ladders and costly signaling. Every forecast is staked in a ledger with a probability, a horizon and a falsifier, following Tetlock's prescription. The engine also holds itself to Jiang's rule: when predictions miss, fix the reasoning, not the story.

The engine went dormant from 4 August to 25 October while the war moved on. Its frozen forecasts therefore form a natural test set, written before the outcomes and unedited since.

The report asks four questions:

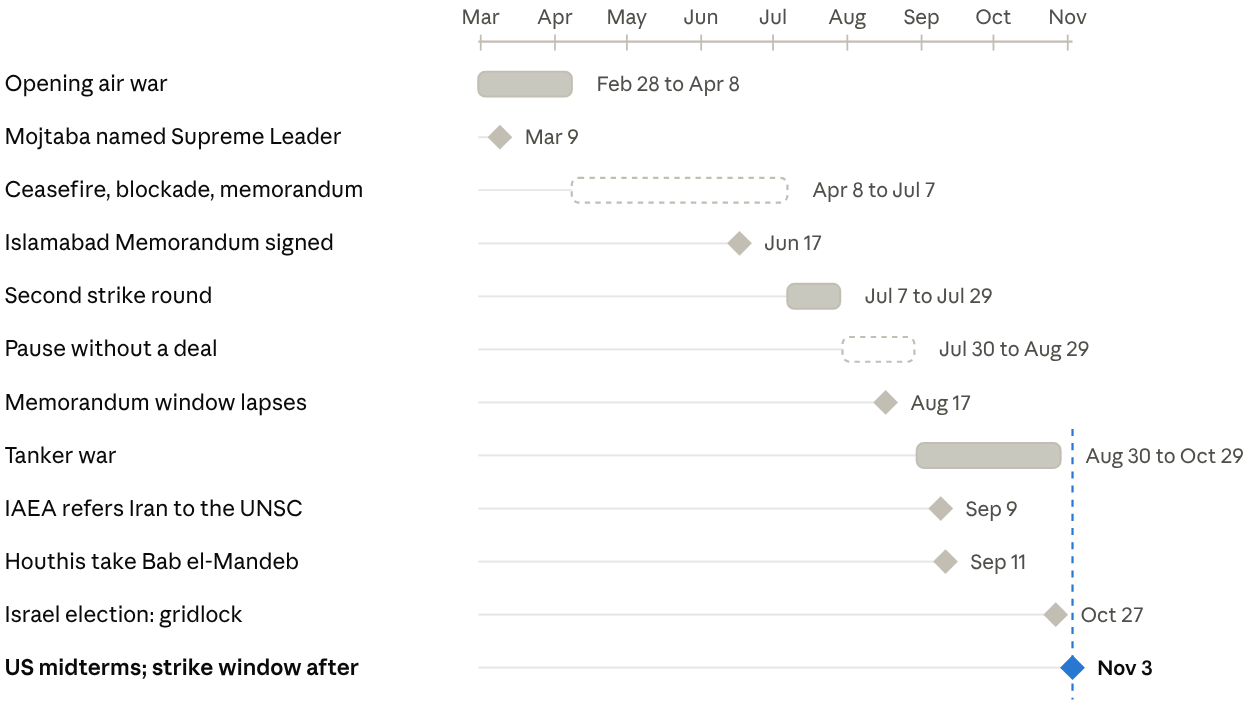
1. **What happened, and why?** What drove the war's arc from February to October, and what game were the players in?
2. **Where was the engine right?** Which geopolitical forecasts held, which failed, and what premise failed in each?
3. **Can ceasefire timing be predicted?** Do survival models on the historical record predict the day a ceasefire fails or a war pauses?
4. **What comes next?** Given the record, how should the branches to 30 November be weighted, and which signposts would move them?

# 2. The war, February–October 2026

The war has gone through two full cycles of fighting and pause and is now in a third round of fighting; neither pause produced a settlement. Since 30 August it has been fought mainly at sea, over the Strait of Hormuz.

**Two pauses, no settlement: the war is now a tanker war over Hormuz**

Phases of fighting (filled) and pause (outlined); diamonds mark turning points, Feb to Nov 2026




War timeline · 5 phases, 7 turning points · dates from the sources in Appendix B

**Opening war (28 Feb – 8 Apr).** US and Israeli strikes ("Epic Fury" and "Roaring Lion") killed Supreme Leader Ali Khamenei on the first day. The Assembly of Experts named his son Mojtaba on 8–9 March as a "unanimous" choice. Reported tallies (59 of 88 votes), a disputed online vote and boycott threats under IRGC pressure suggest otherwise.

**Ceasefire, blockade and memorandum (8 Apr – 7 Jul).** A two-week ceasefire on 8 April was followed by a US naval blockade from 13 April. The Islamabad Memorandum of 17 June set a 14-point, 60-day window to about 16 August. A US sanctions license (General License X, 21–22 June) briefly eased Iranian energy exports.

**Second round (7 – 29 Jul).** Three tankers were hit near Hormuz. The US revoked the license on 7 July, and Trump declared the memorandum "over" on 8 July. The blockade was reinstated on 13 July, and US forces struck Iran for 13 consecutive nights (11–23 July). The Houthis declared a naval blockade of Saudi Arabia on 20 July. The US paused on 24 July at Iran's request, then answered an Iranian missile attack on US forces in Jordan with a heavy wave on 29 July.

**Pause without a deal (30 Jul – 29 Aug).** For about 31 nights there were no US strikes, with Oman and Qatar mediating. Hormuz stayed shut regardless: Kpler counted about 5 vessels a day between 15 July and 23 August, against roughly 130 before the war. The

memorandum's window lapsed on 16–17 August with no deal and no extension.

**Tanker war (30 Aug – present).**

- **The return to strikes.** Limited US strikes resumed on 30 August, followed by a broad wave on 1 September. Iran fired about 25 ballistic missiles at Jordan, and US bases in Bahrain, Iraq and Kuwait were hit. Both sides then destroyed each other's tankers.
- **Diplomatic escalation.** The IAEA Board referred Iran to the UN Security Council on 9 September.
- **The proxy front.** The Houthis seized Perim Island and control of Bab el-Mandeb on 11 September. In October they struck Riyadh's King Khalid airport, killing 13.
- **Diplomacy stalls.** Trump rejected an Iranian seven-day ceasefire proposal on 26 September. Attacks on shipping reached their highest weekly count of the war from 28 September to 5 October.

| Date | Event | Significance |
|---|---|---|
| 29 Oct | IRGC says tankers hit its mines near Khasab and declares the strait "closed"; the US rejects the mine claim | Chokepoint leverage asserted again |
| 27 Oct | Israel's election ends in gridlock: Netanyahu's bloc wins about 53 seats, short of the 61 needed | Caretaker government; narrower Israeli win-set |
| 25 Oct | Trump: "no strike today", weighing "a deal" against "finish them off"; Qatar delivers a draft in Tehran | Deal-or-strike fork opens |
| 22 Oct | DOJ sues to block certification of Alaska's Senate result | US election administration contested |
| 18 Oct | IMF PortWatch counts 4 Hormuz transits (about 85 a day pre-war) | Flows collapse after a September recovery |
| 10 Oct | Houthi missile hits Riyadh airport, 13 killed | Proxy war reaches the Saudi capital |
| 26 Sept | Trump rejects Iran's seven-day ceasefire proposal | Diplomacy stalls |
| 11 Sept | Houthis take Perim and control Bab el-Mandeb | Second chokepoint contested |

| Date | Event | Significance |
|---|---|---|
| 9 Sept | IAEA Board refers Iran to the UN Security Council (23–3–8) | Nuclear track escalates |
| 1 Sept | Broad US strike wave; about 25 Iranian ballistic missiles at Jordan | Tit-for-tat resumes |
| 16–17 Aug | Islamabad window lapses; no deal, no extension | The master clock expires into stasis |
| 30 Jul | Pause begins: about 31 nights without US strikes | Pause without a deal |
| 11–23 Jul | 13 straight nights of US strikes | Second round |
| 8 Jul | Trump declares the memorandum "over" | Framework collapses |
| 17 Jun | Islamabad Memorandum signed; 60-day window | Master clock set |
| 13 Apr | US naval blockade begins | Economic rung of the ladder |
| 8 Apr | Two-week ceasefire | First pause |
| 8–9 Mar | Mojtaba Khamenei named Supreme Leader | Succession settled, contested |
| 28 Feb | War begins; Ali Khamenei killed | Decapitation rung |

Inside Iran, the new Supreme Leader has not made a verified public appearance. He has issued only written statements, an undated clip of under 15 seconds (9 August) and a handwritten letter (released 22 September). Power has visibly shifted toward the security apparatus: a former IRGC commander now heads the Supreme National Security Council, and President Pezeshkian described a six-member decision team on 27 September. The nuclear track worsened in step. Israeli intelligence reported centrifuges moved into the Pickaxe Mountain site (21 July), and the IAEA could not verify about 441 kg of 60%-enriched uranium (early September) before the Board's referral.

# 3. The strategic structure

The war is a constitutive dispute over who sets the terms of passage through Hormuz. Its equilibrium is a pause without a deal, punctuated by tests of resolve, because the chokepoint makes that state cheaper for both sides than either settling or escalating.

## 3.1 Players

Nominal allies are modeled separately, since their best responses diverge.

| Player | Objective | Instruments | Binding constraint |
|---|---|---|---|
| United States (Trump) | Reopen the strait and end the nuclear program on US terms | Air and naval strikes, blockade, sanctions | Midterms on 3 Nov; gasoline above $4 makes a pre-election spike costly; a reversal-prone decider |
| Iran: political leadership | Economic relief without surrender | Negotiation through Qatar, Oman and Pakistan | Cannot deliver the strait: the IRGC controls it |
| Iran: IRGC | Keep the leverage; deter regime change | Mines, tanker attacks, missiles, proxies, nuclear ambiguity | Strike costs; Supreme Leader unseen, so authority is diffuse |
| Israel | Permanent Iranian weakness | Strikes on Iran-linked targets; intelligence | Caretaker government after 27 Oct; US restraint |
| Saudi Arabia and the Gulf | Restore exports; end Houthi attacks | Basing, pipelines, money | Exports squeezed at both ends; a capital under fire |
| Houthis | Legitimacy and leverage | Missiles on shipping and Saudi cities; Bab el-Mandeb | Gain from the war continuing |
| Qatar, Oman, Pakistan | A settlement | Shuttle mediation, drafts | No enforcement power |

Iran′s split sovereignty is the crux. The leadership that can sign a deal cannot open the strait, and the force that can open it has no reason to.

## 3.2 Why no bargain closes

Bargaining theory says war is inefficient, so a settlement both sides prefer exists; it fails only for identifiable reasons. Here, two of Fearon's three mechanisms bind:

- **A commitment problem.** Iran cannot credibly promise to keep the strait open once relief arrives. The US cannot credibly promise relief, or forgo regime change, once the strait reopens. The asset in dispute is the *control* of a chokepoint, which is exactly what each side would have to surrender first.
- **Private information about resolve.** Iran's new leadership is opaque: the Supreme Leader is unseen, and the IRGC is ascendant. The US decider announces and reverses. Neither can read the other's reservation point.

The third mechanism, an indivisible issue, is being *constructed*. Iran's foreign minister says Hormuz "will not return" to its previous state.

Because the dispute is over system architecture rather than a divisible surplus, interim truces are unstable by construction (doctrine §11.5). The record bears this out: two pauses and one memorandum, and none produced a settlement.

## 3.3 A stylized model of the current fork

Take two strategies each: the US either *holds* (blockade, no campaign) or *strikes* (a sustained campaign); Iran either *keeps the strait shut* or *reopens it for relief*. The ordinal payoffs below are the analyst's reading of each side's stated aims and constraints (4 = best). They are not estimates.

| | Iran keeps it shut | Iran reopens for relief |
|---|---|---|
| **US holds** | **US 2, Iran 3** (equilibrium before the midterms) | US 4, Iran 2 |
| **US strikes, before 3 Nov** | US 1, Iran 2 | US 2, Iran 1 |
| **US strikes, after 3 Nov** | US 3, Iran 2 | US 2, Iran 1 |

Before the midterms, holding is the US best response to a closed strait (2 against 1), and keeping it closed is Iran's best response to a holding US (3 against 2). *Hold, shut* is therefore the equilibrium, which is the frozen tanker war observed since August. Iran

never prefers to reopen in this model: a deal requires raising its payoff from reopening. The Qatari draft tries exactly that, trading UN inspections for sanctions relief.

After 3 November the election constraint lifts. If the US payoff from striking a closed strait rises above that of holding, the equilibrium moves to *strike, shut*: the reported post-midterm campaign.

The model does not say whether the flip happens. That turns on one decider's preference ordering, which is precisely what the record shows the engine cannot read (Section 6). The doctrine therefore caps that branch at 0.35 until execution signals appear.

### 3.4 The escalation ladder

The two sides have climbed different ladders without skipping rungs:

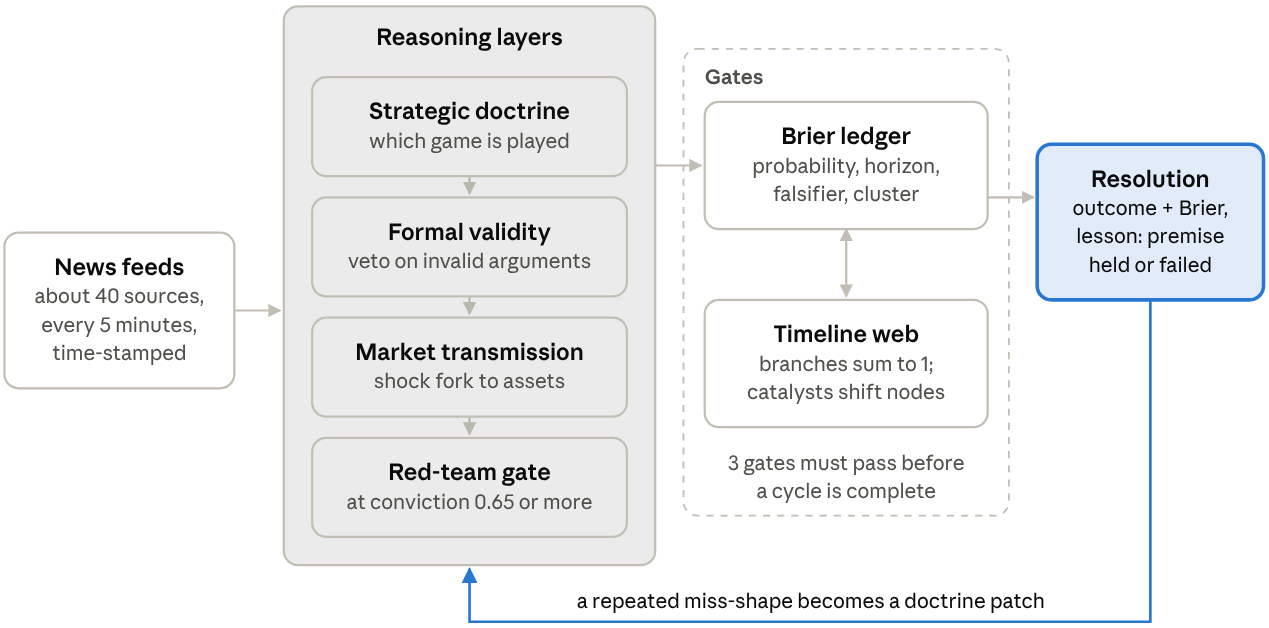
- **The US:** decapitation, then military targets, then an economic blockade. It has repeatedly withheld the next rung, strikes on energy infrastructure (#72 predicted those for 3 August; they did not come).
- **Iran:** missiles at US bases, then mines and tankers, then proxy fire on Saudi civilian sites (Riyadh airport). The desalination rung, which threatens Gulf cities' water, has not been touched (open call #37).

This is escalation *control*, not dominance. Each tanker-for-tanker reprisal is a costly signal of resolve inside a frame that both sides keep below full war.

## 4. The forecasting system

The engine is an accountability machine over expert judgment, not a solver. It computes coherence, conditional-path probabilities and Brier scores; the probabilities themselves are analyst estimates stored as data.

**Every forecast is staked, scored and fed back into the doctrine**




System architecture · 4 reasoning layers, 2 records, 3 gates, 1 learning loop

Forecasts move left to right. The highlighted loop is the learning rule: every resolution files a lesson, and a miss-shape that repeats forces a doctrine patch.

**Ingestion.** About 40 RSS and JSON feeds are polled every five minutes. They cover wires, central banks, defense, energy, shipping, OSINT and principal rhetoric. Items are de-duplicated and stamped with their *reported* time, so staleness is visible. Watchlist matches can trigger a headless analysis; this ships in dry-run mode.

**Reasoning layers.** Each analysis passes through four skills in a fixed order of authority:

| Layer | Role | Representative rules |
|---|---|---|
| Strategic doctrine | Which game is being played | Chessboard actor decomposition; constitutive vs distributive conflict (§11.5); signal-vs-strategy filter (§12.1); escalation–mediation oscillation (§16.12); hard cap on strike-planning reports (§16.14) |

| Layer | Role | Representative rules |
|---|---|---|
| Formal game theory | Whether the argument is valid; wins every validity conflict | Fearon mechanisms, costly signaling, Kahn ladders, prospect theory, global games, Crawford–Sobel coarseness; a 25-item checklist |
| Market transmission | How the event transmits through energy, trade and finance | Supply / generic / demand shock fork (Kilian; Caldara–Iacoviello); chokepoint and corridor channels; horizon matching |
| Contrarian red team | Attack the book before it is staked | A six-step gate (assumptions, ACH, premortem, cascade check, reference class, extremizing) for any call at conviction ⩾ 0.65 |

**Conviction rubric.** Each event gets a conviction score in [0, 1], the sum of five anchored components: source quality (0.20), capital committed (0.30), formal validity (0.20), multi-lens convergence (0.15) and precondition status (0.15). Calls below 0.35 are noise and are not ledgered.

**The ledger.** An SQLite table holds every prediction with its probability, horizon, falsifier, sensitivity tag, conviction, tickers, direction and correlation cluster. Resolution records a binary outcome and its Brier score. A separate table requires a *lesson* for every resolution: a verdict (hit-clean, hit-wrong-reason, partial or miss), the premise that held or failed, and the doctrine patch it forced.

**The timeline web.** A time-laned directed acyclic graph of branching futures, in which every ledger row must be wired to a node. Sibling branches must sum to about 1. Scheduled catalysts must carry "shift" edges that state which downstream probabilities their outcome moves.

**Gates.** A cycle is incomplete until three checks pass: every prediction is wired to the map with no dead-end catalysts; nothing is overdue-unscored or resolved without a lesson; and a full system debug runs clean (syntax, database, JSON, branch sums, dashboard routes).

# 5. Data and scoring

Every result here is scored against an outcome dated after the forecast and recorded in a committed file.

## 3.1 Data

| Source | Content | Used in |
|---|---|---|
| Ledger (`ledger.db`) | 81 predictions logged 21 Apr – 25 Oct 2026, with falsifiers and lessons; 19 resolved geopolitical calls | Studies 1, 3 |
| Fact-finding passes, 25 and 29 Oct | Dated, sourced answers on the Iran war, Hormuz, OPEC+, the IAEA, US election administration and Israel's election | Studies 1, 3; Section 2 |
| News ingest | About 40 feeds, 773 new items on 29 Oct, each with its reported time | Study 3 |
| PA-X v10 (Edinburgh) | 2,257 peace agreements 1990–2025, 483 at the ceasefire stage, linked to UCDP conflict ids | Study 2 |
| UCDP GED 26.1 + candidate 26.0.1–9 | 434,601 georeferenced events, 1989 – Sept 2026 | Study 2 |
| UCDP Conflict Termination v4 | Interstate typing | Study 2 |
| | | |

## 3.2 Point-in-time discipline

Look-ahead is the dominant threat in any retrospective forecast study. It is controlled at three levels:

1. **Freezing.** Ledger predictions were written before their outcomes and never edited. Study 1 only scores them.
2. **Censoring.** In Study 2 each walk-forward fit sees only cases signed before 1 January of the test year. Every training outcome is right-censored at that cutoff, so a case still running on 31 December counts as censored.

3. **Forward only.** Study 3 scores calls logged on 25 October against events after that date, sourced at the time they occurred.

### 3.3 Scoring

For a binary event with forecast probability p and outcome o ∈ {0, 1}, the Brier score is the mean squared error (lower is better; a constant 0.5 forecast scores 0.25):

```
\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N} (p_i - o_i)^2
```

Murphy's decomposition separates calibration from discrimination. Forecasts are binned into K groups of size n_k, with mean forecast p_k, observed frequency ō_k and overall base rate ō:

```
\mathrm{BS} = \underbrace{\tfrac{1}{N}\sum_k n_k (p_k - \bar
o_k)^2}_{\text{reliability}} \; - \; \underbrace{\tfrac{1}{N}\sum_k n_k (\bar
o_k - \bar o)^2}_{\text{resolution}} \; + \; \underbrace{\bar o (1 - \bar
o)}_{\text{uncertainty}}
```

Ranking quality in the survival models is Harrell's C, the share of comparable pairs in which the case that failed first had the shorter predicted median. For timing, the point forecast is the median of the predictive distribution, which minimizes expected absolute error in days.

**Dependence.** Calls on the same war share a regime: if the war-state model is wrong, many rows fail together. The ledger tags each call with a correlation cluster, and 16 of the 19 geopolitical calls sit in one (the Iran war). Results are therefore reported by mechanism family, and no significance claim rests on raw call counts.

## 6. Study 1: the geopolitical forecast record

The engine's 19 resolved geopolitical forecasts scored 12 correct (63%), with Brier 0.236 against 0.25 for a coin. The record splits sharply: it read the war's physical and structural state almost perfectly and its principals' decisions almost never.

### 6.1 Scope and procedure

The subset contains every resolved call about events rather than prices: war and ceasefire, chokepoint transit, regime and succession, proxy and Israeli action, the nuclear program, OPEC+ policy and US election administration. Calls on oil prices, inflation,

central banks or equities are excluded by design. The full ledger, those included, stands at 35/53.

Fifteen of the 19 were resolved at relaunch, 14–79 days past horizon, from sourced fact-finding passes. Five falsifiers in the ledger were defective, two of them in this subset (#68, #14). Defective contracts were scored on the prediction's substance, and both resolved as misses, the conservative direction.

## 6.2 Results by mechanism

On this subset the Murphy decomposition gives reliability 0.032, resolution 0.031 and uncertainty 0.233. Resolution is nearly four times the full ledger's (0.008): the engine discriminates far better on geopolitical events than on markets.

The grouping below was made after seeing outcomes, so it is diagnostic, not a test:

| Mechanism family | Calls | Correct | Brier | Mean stated P |
|---|---|---|---|---|
| Physical and structural state | 9 | **9** | 0.13 | 0.66 |
| US political instruments | 2 | 2 | 0.15 | 0.63 |
| A principal's discrete decision: strike, stop, sign, cut output | 7 | **1** | 0.38 | 0.64 |
| Stale structural premise | 1 | 0 | 0.38 | 0.62 |

The engine's stated confidence was the same across families (0.62–0.66). It did not know where its skill was.

## 6.3 The calls

| # | Forecast (abridged) | P | Result | Evidence |
|---|---|---|---|---|
| 73 | Hormuz does not reach normalized commercial transit by 7 Aug | 0.85 | Hit | Kpler about 5 vessels a day, 15 Jul – 23 Aug, vs about 130 pre-war |
| 22 | Hormuz not normalized within 60 days (transit or war-risk premia) | 0.68 | Hit | War-risk cover 3–10% of hull value vs 0.15–0.25% pre-war |

| # | Forecast (abridged) | P | Result | Evidence |
|---|---|---|---|---|
| 58 | Saudi seaborne crude exports fall more than 20% in August | 0.55 | Hit | Tanker tracking: −22% to −33% (self-reported JODI data disagree) |
| 15 | A further maritime or proxy attack within 60 days | 0.70 | Hit | Many: Red Sea, Bab el-Mandeb and Hormuz, July–September |
| 17 | The Islamic Republic does not collapse within 60 days | 0.88 | Hit | Government intact; president at the SCO summit 31 Aug |
| 18 | Mojtaba stays low-visibility, sharing power with the IRGC | 0.66 | Hit | No verified appearance; former IRGC chief heads the Supreme National Security Council |
| 20 | Iran or a proxy takes vengeance-framed action within 60 days | 0.55 | Hit | Reported plot against Trump; bounty on US soldiers (16 Aug) |
| 21 | Israel strikes an Iran-linked target within 60 days | 0.58 | Hit | Strikes on Hezbollah, 15 Aug and 6 Sept |
| 30 | Credible reporting of nuclear reconstitution within 90 days | 0.52 | Hit | Pickaxe Mountain report; IAEA referral 9 Sept |
| 48 | The 16 Jul address frames elections as compromised and pledges action | 0.70 | Hit | Content as described |
| 49 | A concrete federal election instrument within 30 days | 0.55 | Hit | DHS letters to four states (17 Jul); Supreme Court application (27 Jul) |
| 64 | The US strikes another nuclear-associated site by 27 Jul | 0.62 | Hit | Unfinished plant struck during the 13-night campaign |
| 19 | No return to war-scale conflict within 60 days | 0.70 | Miss | US strikes resumed three days after logging |
| 59 | The 10-day ceasefire attempt fails | 0.72 | Miss | About 31 nights without US strikes from 30 Jul |
| 63 | US strikes the Houthis in Yemen and announces an escort framework | 0.75 | Miss | Neither; Washington chose diplomatic pressure on Iran |

| # | Forecast (abridged) | P | Result | Evidence |
|---|---|---|---|---|
| 72 | US or Israel strike Iranian energy infrastructure by 3 Aug | 0.55 | Miss | Trump stood down on 1 Aug, citing "perimeters of a deal" |
| 68 | A mediation off-ramp by the 16 Aug window | 0.55 | Miss | No deal; window lapsed; strait still shut |
| 29 | OPEC+ pauses its output hikes at the next meeting | 0.58 | Miss | Hikes continued on 2 Aug; pauses came only in September |
| 14 | Supreme-leader succession stays unresolved for three months | 0.62 | Miss | Succession settled on 9 Mar, before the call was logged |

### 6.4 Why the misses happened

Every miss is traced to a named premise in the ledger's lesson table:

- **Reading leaks and rhetoric as decisions (#63, #72, #59).** A leaked strike plan was logged at 0.55 even though the engine's own sensitivity tag named Trump's reversal pattern as the brake. The fix is a hard cap of 0.35 on strike-planning reports until execution signals appear (doctrine §16.14).
- **Confusing a pause with an off-ramp (#68).** The engine priced a mediated settlement at 0.55. What came was the branch it had priced at 0.07, a pause without a deal. Pauses, deals and reopenings are now separate branches (§16.12b), and the call's falsifier tested a rival scenario instead of its own negation (formal §8.1).
- **Stale premises at logging (#19, #14, #29).** #19 was logged the day the memorandum was declared over. #14 treated a settled succession as live. #29 assumed a "normalizing-Hormuz glut" that the blockade's return ended days later. Premises must now be re-verified against a primary source when a call is logged (§16.13a).

The hits share a different property. Each turned on a measured state (vessel counts, insurance premia, the regime's continuity, the leader's visibility) or on a base rate of proxy behavior. None required reading one person's mind.

## 7. Study 2: predicting ceasefire timing

Survival models on the historical record rank which ceasefires fail first (C-index 0.83). They predict the failure *day* only for ceasefires that collapse within three days.

## 5.1 Case construction

Ceasefire-stage agreements in PA-X v10 were linked to UCDP conflict ids; agreements on the same conflict within three days were merged into one event. Daily state-based fatalities came from UCDP GED, using only events dated to the day or week. Each ceasefire gets two outcomes:

- **First fatal violation (v1):** the first day after signing with any fatality.
- **Return to war (r25):** the first day the trailing seven-day toll reaches 25.

A case still running on 30 September 2026 is censored. This yields **401 ceasefires**, of which 337 returned to war by the cutoff. For the reverse question, **1,380 war spells** were built: an onset, or a resumption, after a lull of more than 60 days. Of these, 175 reached a ceasefire.

Predictors are all dated before signing:

- violence in the prior 7, 30, 90 and 365 days, and its trend;
- conflict age;
- interstate, local, unilateral, mixed and third-party flags;
- the count of prior ceasefires and prior breakdowns *known at signing*;
- the duration of the last breakdown.

## 5.2 Models

Four estimators produce a full survival curve S(t) over days 0–3,650:

- a Kaplan–Meier outside view;
- an interval-censored log-normal accelerated-failure-time (AFT) model;
- a reference class: the weighted Kaplan–Meier curve of the 30 nearest past cases;
- an equal-weight ensemble of the AFT model and the reference class.

In the AFT model, an event recorded on day t means failure in (t−1, t]. With location $\mu = x^{\top}\beta$, scale σ and Φ the standard normal CDF, the likelihood is:

```
\mathcal{L} = \prod_{i \in E} \left[ \Phi\!\left(\tfrac{\ln t_i - \mu_i}
{\sigma}\right) - \Phi\!\left(\tfrac{\ln (t_i - 1) - \mu_i}{\sigma}\right)
\right] \; \prod_{j \in C} \left[ 1 - \Phi\!\left(\tfrac{\ln c_j - \mu_j}
{\sigma}\right) \right]
```

It is fit by L2-penalized gradient ascent. On synthetic log-normal data it recovers the truth: intercept 2.90 (true 3.00), slope 0.99 (1.00), σ 1.03 (1.00).

A second, **day-by-day model** re-forecasts at the end of every ceasefire day. It is a logistic hazard on the static predictors plus seven running features (the day count; deaths in the last 1, 3 and 7 days; cumulative deaths; days since the last fatality; whether any fatality has occurred), estimated for one-day and seven-day horizons.

## 5.3 Results (walk-forward, 2000–2026)

| Return to war (267 test ceasefires) | Kaplan–Meier | Ensemble |
|---|---|---|
| Exact day | 0.0% | 8.1% |
| Within ±1 day | 0.0% | 17.5% |
| Within ±7 days | 0.9% | 25.6% |
| Within ±30 days | 7.6% | 37.2% |
| Median absolute error | 103 days | 76 days |
| C-index | 0.44 | **0.83** |
| Brier, P(war within 30 days) | 0.231 | **0.093** |

Accuracy depends almost entirely on how long the ceasefire actually lasted:

| Actual duration | Cases | Within ±1 day | Within ±7 days | Median error |
|---|---|---|---|---|
| 1–3 days | 55 | **62%** | **82%** | 1 day |
| 4–30 days | 37 | 11% | 30% | 24 days |
| 31–365 days | 79 | 1% | 1% | 165 days (naive: 62) |
| Over a year | 52 | 0% | 0% | 962 days |

For the first fatal violation, the ensemble is exact on 26% of cases and within ±7 days on 43% (median error 10 days; naive 1%, 28%, 13 days). For war spells reaching a ceasefire, it ranks well (C 0.82) and roughly halves the one-year Brier score (0.21 vs 0.39). Its timing is poor: 2.8% of calls fall within ±7 days. With only 43 interstate spells in the record, 8 of which reached a ceasefire, that direction is unreliable for interstate wars.

The day-by-day model ranks dangerous days well. Its AUC is 0.917 for war the next day and 0.873 for war within seven days, against 0.710 and 0.664 for the baseline hazard. It still warns late and rarely:

| Alarm threshold, P(war within 7 days) | 0.5 | 0.3 |
|---|---|---|
| Collapses warned within the prior 7 days | 16% | 32% |
| Warned at least 2 days ahead | 9% | 23% |
| Median lead | 2 days | 3 days |
| Alarm precision | 60% | 40% |

The median P(war tomorrow) on the eve of an actual collapse was 4.4%.

## 5.4 The 2026 Iran holdout

| Ceasefire | Call at signing | Actual | Error |
|---|---|---|---|
| 8 Apr: first violation | day 1 | day 1 | exact |
| 17 Jun: first violation | day 17 | day 10 | 7 days |
| 25 Jul: first violation | day 3 | day 5 | 2 days |
| 8 Apr: return to war | day 11 | day 95 | 84 days |
| 17 Jun: return to war | day 157 | day 25 | 132 days |

Replaying the 17 June memorandum day by day, P(war within 7 days) rose to 0.47–0.53 on 8–9 July. War registered on 12 July, three to four days later. Under the 8 April ceasefire, the model raised a false alarm on 1 May.

GED's fatality counts understate a war fought from the air and at sea. Thirteen nights of US strikes from 11 July register at 2–8 deaths a day, so r25 is a weak proxy for "return to war" in this conflict.

## 8. Study 3: the relaunch calls, tracked forward

Four days after relaunch, both live geopolitical calls are on track, and the scenario weights moved as the evidence predicted. These are the only fully out-of-sample results in the report.

| # | Call (logged 25 Oct) | P | Status, 29 Oct | Evidence |
|---|---|---|---|---|
| 79 | No US strike on Iranian land territory, 26 Oct – 3 Nov | 0.70 | On track | None found through 29 Oct. The US warned Gulf governments to prepare for strikes *after* the midterms, the sequencing the call bet on |
| 56 | Midterms held, with at least one federal certification fight | 0.55 | On track | DOJ suit to block certification of Alaska's Senate result pending |

#79 shows the strike-planning cap working as designed. On 25 October, CENTCOM's plans, Israeli Air Force activity and Trump's "finish them off" rhetoric all pointed to imminent strikes. The engine weighted the stated post-midterm sequencing and the pre-election cost of an oil spike, not the leaks.

Three catalysts on the engine's board resolved:

- **Israel's election (27 Oct)** ended in gridlock. Netanyahu's bloc won about 53 seats, short of the 61 needed, and a caretaker government is likely. The feared Israeli first strike around the vote, #79's named risk, did not occur. A caretaker prime minister at risk of losing power is in the loss domain of prospect theory, so the risk of a unilateral gamble has not disappeared.
- **Qatar's mediation** produced a seven-day confidence-building draft (delivered 25 Oct) that would trade UN inspections of bombed sites for sanctions relief. A senior Iranian official told Reuters there was no new proposal and the ideas were unacceptable.
- **Hormuz** tightened again. On 29 October the IRGC said several tankers hit its mines near Khasab and declared the strait "closed". The US rejected the mine claim, and UKMTO reported a vessel fire near Khasab.

The scenario weights moved accordingly:

| Branch to 30 Nov | 25 Oct | 29 Oct | Why |
|---|---|---|---|
| Frozen tanker war continues | 0.50 | **0.55** | Mines, closure claim, no deal |
| Post-midterm US strike campaign | 0.35 | **0.35** | Held at the cap: the Gulf warning is closer to execution, but no ordnance or flight bans yet |
| Deal confirmed by both sides | 0.15 | **0.10** | Tehran rejected the Qatari draft |

# 9. Outlook

The most likely path through 30 November is a continued tanker war, with the main risk a US strike campaign in the weeks after 3 November. A deal is the least likely branch, and even a signed one would probably see fighting resume within a quarter.

## 9.1 Signposts

Each branch has observable markers. Following the doctrine's own rule, execution signals count; statements do not.

| Branch (weights in Section 8) | Would be confirmed by | Would be undercut by |
|---|---|---|
| Frozen tanker war | Further mine and tanker incidents; Hormuz transits below about 20 a day (Kpler, IMF PortWatch); war-risk premia above 5% of hull value | Transits recovering above half of pre-war |
| Post-midterm US strike campaign | Flight bans over the Gulf; carrier or bomber movements; embassy drawdowns; a mass-casualty Iranian strike on US forces | A public Iranian move on the Qatari draft; Trump citing "perimeters of a deal" again |

| Branch (weights in Section 8) | Would be confirmed by | Would be undercut by |
|---|---|---|
| Deal confirmed by both sides | IRGC endorsement, not only the foreign ministry's; UN inspectors at bombed sites; a US sanctions license; physical reopening | Another "surrender" framing from Tehran; renewed mining |

The engine's record argues for weighting the right-hand column of the deal row heavily. Twice before ("perimeters of a deal" on 1 August, the Hormuz framework in early August), deal talk did not become reopened traffic.

## 9.2 If a deal is signed

The ceasefire-timing engine (Section 7) gives a conditional read for a ceasefire signed around 1 November:

| Outcome | Median | Within 7 days | Within 30 days | Within 90 days |
|---|---|---|---|---|
| First fatal violation | day 12 | 39% | 75% | — |
| Return to war (25 deaths in a week) | day 93 | 7% | 24% | 50% |

This is the historical base rate for a ceasefire in a conflict with this violence history and prior breakdowns. Its reverse direction, how long until a war reaches a ceasefire, gave about 5–11% for a ceasefire by 30 November. That direction missed both 2026 Iran ceasefires badly, so it is noted, not relied on.

## 9.3 Longer-horizon calls still open

| # | Forecast | P | Horizon | Status, 29 Oct |
|---|---|---|---|---|
| 38 | US ground invasion of Iran (Jiang's call) | 0.70 | 31 Dec 2026 | At risk: the plans reported are air and naval, not ground |
| 53 | No PLA kinetic action against Taiwan in 2026 | 0.93 | 31 Dec 2026 | On track |

| # | Forecast | P | Horizon | Status, 29 Oct |
|---|---|---|---|---|
| 12 | No enforced, published Hormuz transit-toll schedule | 0.62 | 4 Jan 2027 | On track, but Iran asserts permission-based passage |
| 16 | Attributed Israeli action on Iranian territory or assets | 0.74 | 8 Jan 2027 | At risk: none on Iran proper found since 8 Jun |
| 36 | US ground troops, or a compensated cessation deal | 0.45 | 13 Jan 2027 | At risk |
| 10 | No durable US–Iran peace within 12 months of the memorandum | 0.70 | 4 Jul 2027 | On track |
| 35 | The war institutionalizes as a frozen conflict | 0.55 | 13 Jul 2027 | On track |
| 37 | Attributed strike or plot against Gulf desalination | 0.40 | 13 Jul 2027 | Open: the rung is untouched |
| 1 | The war continues for years despite ceasefires (Jiang's call) | 0.70 | 21 Apr 2028 | On track |
| 34 | Iranian nuclear test or declared weapons capability | 0.35 | 13 Jul 2028 | Open: IAEA referral raises the stakes |

The at-risk rows (#38, #16, #36) all bet on discrete escalation decisions, the family the engine reads worst. If they fail as they look set to, the projected geopolitical record falls from 12/19 to 12/22.

## 10. Discussion

The record supports one claim above the rest: in this war, constraints were forecastable and choices were not. A strategic doctrine earns its keep by identifying the constraints and the equilibrium they produce, not by predicting what a leader will decide on a given night.

### 10.1 Constraints versus choices

Three independent bodies of evidence point the same way:

- **Study 1.** Physical and structural calls went 9 for 9; decision calls went 1 for 7.

- **Study 2.** The ceasefire engine ranks collapse risk well (C = 0.83), because risk is written in structure: violence before signing, prior breakdowns, conflict age. It cannot date a collapse: on the eve of an actual one, its median probability of war the next day was 4%.
- **Study 3.** The live call that holds up (#79) works by *refusing* to forecast the decision. It bets on a stated constraint, the election calendar, instead.

The theory explains why. A decision by a single principal turns on private information and a preference ordering that public aggregates do not contain. By Crawford–Sobel logic, what such a principal says publicly carries only coarse information. Structure, by contrast, is observable and slow to change.

## 10.2 The oscillation was an equilibrium, not noise

The engine's doctrine correctly said the war would oscillate rather than climb a ladder. It drew the wrong conclusion: that oscillation implied a mediated exit. Section 3's model shows why the pause without a deal is self-enforcing. As long as the strait is shut, holding is cheap for the US and keeping the strait shut is Iran's best response, so no one needs to sign anything.

That reframes the forecasting question. The useful question is not "will they make peace?" but "what would change the payoffs?" The candidates are concrete: the election constraint lifting, relief large enough to outbid the strait's leverage, or a mass-casualty shock.

## 10.3 The physical layer outranks the announcement layer

The same week produced a hit and a miss on the same subject. #73 ("Hormuz will not normalize") held because it read vessel counts. #68 ("a mediated off-ramp") failed because it read frameworks and talks. Interested parties' statements about a chokepoint are cheap talk. Transit counts and insurance premia are costly, revealed behavior.

## 10.4 Leaks are instruments

Schelling's threat that leaves something to chance is the US posture throughout: plans leaked, sequenced and withdrawn. A leaked plan from a principal who reverses is a bargaining move, so its probability as a forecast should be capped. The engine learned this the hard way (#72 in August). It applied the lesson in October (#79).

### 10.5 Confidence should depend on the mechanism

The engine stated about the same confidence (0.62–0.66) whether a call turned on a vessel count or on a leader's mood. The remedy is a mechanism-type prior:

- **Measured-state calls:** confidence can rise.
- **Decision calls:** shrink toward the base rate unless an execution signal is observed.

Had the seven decision calls been stated at the 0.5 base rate, their Brier score would have been 0.25 instead of 0.38, with the measured-state calls untouched.

### 10.6 What would falsify this reading

Two outcomes by 30 November would show the equilibrium story wrong:

- a deal with *physical* reopening, transits above half of pre-war levels;
- a full air campaign *before* the midterms.

Either would mean the payoffs were misread, not merely that a decider surprised.

## 11. Threats to validity

The largest threats are small, correlated samples and a grouping chosen after the fact. Each is bounded below; none is removable.

| Threat | Severity | What limits it, and what remains |
|---|---|---|
| Small, correlated samples | High | 19 geopolitical calls, 16 of them in one conflict cluster. The 9/9 and 1/7 contrasts are striking but not statistically secure |
| Post hoc grouping | High | The mechanism families were drawn after outcomes were known. The contrast is offered as a hypothesis for forward testing (Section 12), not a result |
| Scope choice | Medium | Price and market calls are excluded by design. The full ledger record (35/53) is stated alongside so the subset cannot flatter |
| Scoring discretion | Low | Two defective falsifiers were scored on substance, both against the engine |

| Threat | Severity | What limits it, and what remains |
|---|---|---|
| Fact quality | Medium | Several facts rest on secondary compilations or search excerpts where primary pages refused access. Sources conflict on some points (Saudi exports: tanker tracking vs self-reported data); the falsifier's named source governs |
| Measurement of "war" | Medium | UCDP's fatality counts understate an air and sea war: 13 strike nights register at 2–8 deaths a day. PA-X omits many unilateral and failed ceasefires |
| Stylized model | Medium | Section 3's payoffs are ordinal judgments from stated aims, not estimates. They explain the observed equilibrium; they do not prove it |
| Single conflict | High | One war dominates the 2026 evidence; generalization rests on Study 2's historical record alone |

## 12. Conclusions and future work

The 2026 US–Iran war is a constitutive struggle over a chokepoint whose equilibrium, for now, is a pause without a deal. A game-theoretic engine forecasts that structure well and the principals' choices poorly; it should say so in its probabilities.

**Answers to the research questions:**

1. **What happened, and why?** Two cycles of fighting and pause, then a tanker war. A commitment problem over control of Hormuz, with the strait in the hands of the faction that cannot sign, makes stasis the equilibrium (Sections 2–3).
2. **Where was the engine right?** On the physical and structural state of the war (9/9), and wrong on single decisions (1/7). Its 19 resolved geopolitical calls scored 12 correct, Brier 0.236 (Section 6).
3. **Can ceasefire timing be predicted?** Risk can be ranked (C = 0.83). The day can be named only for ceasefires that fail at once. Adding mediation activity helps marginally (day-by-day AUC 0.873 to 0.877) (Section 7).
4. **What comes next?** A continued tanker war is the modal path to 30 November. The post-midterm strike window is the main risk and is held at the cap until execution signals appear (Sections 8–9).

**Follow-on work completed on 29 October:**

- [x] ~~Mediation precursors added to the ceasefire engine (PA-X agreements, static and daily). Deadlines, flight bans and force posture are hand-coded for the 2026 Iran ceasefires and logged prospectively from the news feed, to become training data.~~
- [x] ~~An ingestion path for the ETH/PRIO ceasefire dataset, and a drafted access request.~~
- [x] ~~An abstention zone in the ledger: mechanical calls between 0.40 and 0.60 are refused, and resolution is tracked before and after the rule (baseline 0.008).~~
- [x] ~~A dormancy alarm (the scoring gate fails after seven idle days) and a daily scheduled scoring session.~~

**Still to do:**

- [ ] Send the ETH/PRIO data request (it needs a person to send it).
- [ ] Add a mechanism-type prior to the rubric: higher confidence for measured-state calls, shrinkage toward the base rate for decision calls (Section 10.5).
- [ ] Test the 9/9 versus 1/7 contrast forward: tag each new geopolitical call's mechanism family when it is logged, and score the families separately.

# Appendix A: Reproducibility

All code, data extracts, ledger and reports live in the private repository `truchard0/GT-Analyst`, on branch `claude/gracious-lamport-6wef9u` (not yet merged to `main`).

| Commit | Content |
|---|---|
| `d2cc1fc` | Handoff document: the scheduled scoring sessions |
| `48ad3f8` | Ceasefire precursors, abstention zone, dormancy alarm, ETH/PRIO hook and request draft |
| `a22ef87` | Study 3: relaunch calls tracked forward; scenario weights re-set |
| `cc02cc0` | Relaunch analysis: calls #79 and #56 wired; the October fork added to the timeline |
| `c69f2cb` | Study 2: ceasefire-timing engine and case base |
| `937d63b` | Study 1: overdue calls scored, lessons and doctrine patches |

```
python pipeline/ledger.py report                                  # calibration,
Murphy decomposition, abstention tracking
python tools/build_ceasefire_data.py                              # Study 2:
download + build the case base
python pipeline/ceasefire_engine.py backtest --target r25     # also v1, cf;
--no-precursors for the 25 Oct engine
python pipeline/ceasefire_engine.py backtest-daily --tau 0.3
python pipeline/ceasefire_engine.py live                          # Iran 2026
replay with precursor annotations
python tools/precursor_log.py summary --days 14
bash tools/daily_cycle.sh                                         # the daily
scoring gate
```

# Appendix B: References

**Forecasting and scoring.**


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**Data.**

- PA-X Peace Agreements Database, v10 (University of Edinburgh).
- UCDP Georeferenced Event Dataset 26.1, candidate GED and Conflict Termination v4 (Uppsala University).
- ETH/PRIO Civil Conflict Ceasefire dataset, described by Clayton et al. (2023), *Journal of Conflict Resolution* 67(7–8); not publicly downloadable.
- Hormuz transit counts from Kpler and IMF PortWatch, as reported in the event sources below.

**Event sources (selected).**

*Chronology and war state*

- Timeline of the 2026 Iran war (secondary compilation, cross-checked)
- 2026 Iranian leadership crisis

- Foley: OFAC rescinds Iranian sanctions relaxation
- Politis: Iran to halt strikes as long as the US pause holds (27 Jul)
- CNN live coverage, 29 Jul
- AFP via The Standard: US strikes Iran for the first time in a month (30 Aug)
- Al Jazeera: the memorandum is set to expire (16 Aug)
- Iran International, 17 Aug
- September 2026 US strikes on Iran
- CNBC: US strikes Iran after Hormuz shipping attacks (1 Sept)
- USNI News: US and Iran trade strikes in Hormuz (4 Sept)
- Korea Times: Iran says resumption of war would not change its stance (25 Oct)

*Hormuz and the Red Sea*

- Al Jazeera: a 95% drop in Hormuz traffic (27 Aug)
- CNBC: Hormuz ship traffic (12 Aug)
- Al Jazeera: Hormuz data vs US claims (3 Sept)
- The National: war-risk premiums surge (17 Jul)
- Oman Observer: Hormuz transits at a two-month low
- Al Jazeera: attacks soar, flows crash again (14 Oct)
- Al Jazeera: ADNOC tanker targeted (8 Aug)
- Long War Journal: Houthis escalate in the Red Sea (6 Aug)
- Euronews: Houthis strike a Saudi tanker (24 Aug)
- Al Jazeera: Houthi missile targets Riyadh (12 Oct)
- Zawya: Mideast oil exports and Saudi shipments
- Arab News: OPEC+ continues hikes (2 Aug)
- Intellinews: Iran says tankers hit mines; US denies

*Iran's leadership, proxies and the nuclear track*

- Jerusalem Post: Mojtaba clip (Aug)
- IBTimes: Mojtaba letter and public appearance (28 Sept)
- 2026 alleged Iranian plot to assassinate Donald Trump
- Gulf News: bounty on US soldiers
- NBC News: Israeli strike in southern Lebanon (15 Aug)
- Al-Monitor: Israeli strikes in southern Lebanon (6 Sept)

- Gulf News: construction at Pickaxe Mountain
- Khaleej Times: IAEA Board refers Iran to the UN Security Council

*The October fork*

- Times of Israel: "no strike today" (25 Oct)
- Iran International: Qatar pushes for a deal (25 Oct)
- CNBC: deal or "finish it off" (26 Oct)
- Gulf News: US and Israel plan week-long strikes
- HNGN: Tehran denies the Qatar plan (28 Oct)
- The War Zone: on the brink of Epic Fury round two (23 Oct)
- i24: Israel's election set for 27 Oct
- Ynet: Israel election results
- Fox News: Netanyahu falls short of a majority
- Al Jazeera: Israel's election, candidates and results (29 Oct)

*US election administration*

- Democracy Docket: DHS letters to state election officials
- SCOTUSblog: mail-ballot appeal
- Al Jazeera: DOJ sues California (15 Oct)
- Alaska Beacon: DOJ sues over Alaska's Senate result (23 Oct)